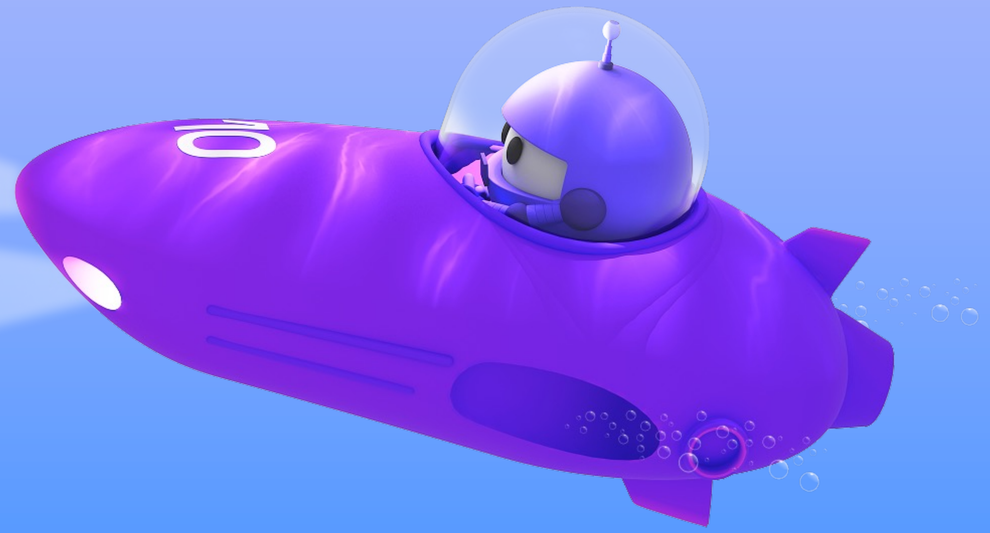
.NET MAUI
What's New in .NET 10

David Ortinau
Principal Product Manager
.NET MAUI



Investments for .NET 10

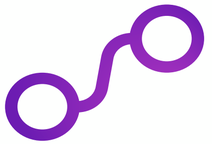
.NET MAUI



Quality



Performance



Simplicity



Modern

A Year of Growth and Acceleration

.NET MAUI



+24%

Developers



-34%

Issues Created



+40%

PRs Created



+34%

PRs Merged


Thank you contributors!

Component Vendors









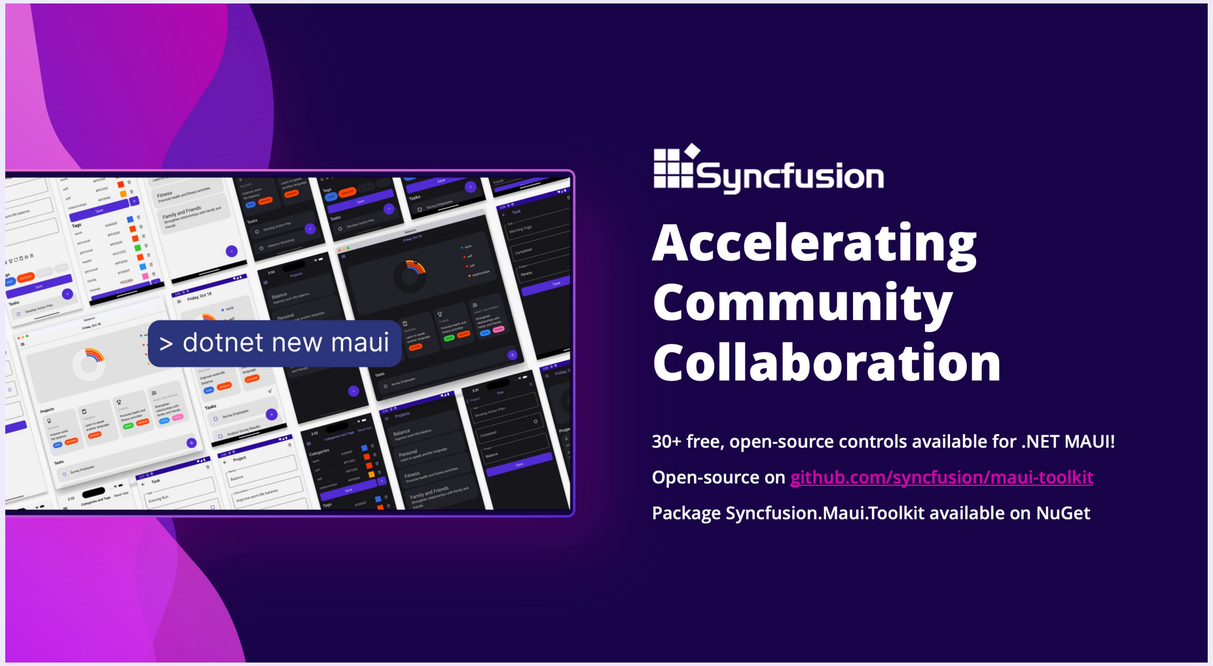


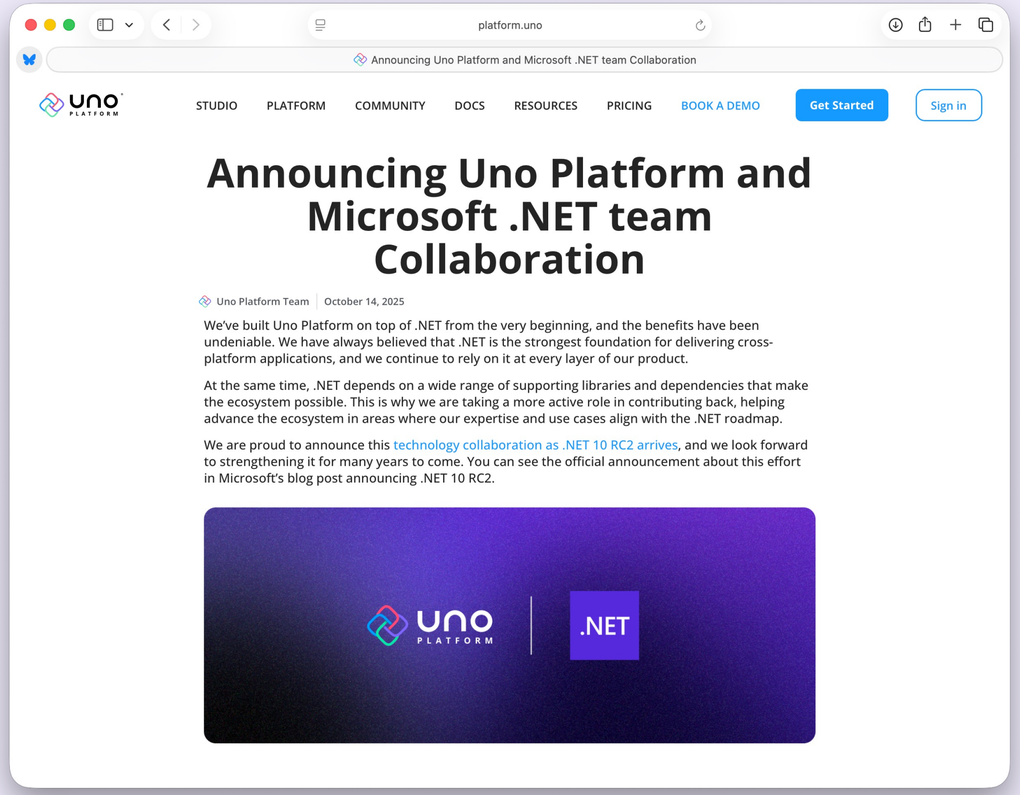






Acceleration
Through Collaboration

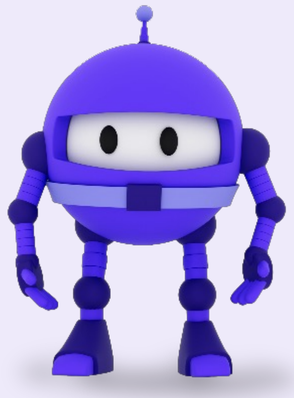




> We have always believed that .NET is the strongest foundation for delivering cross-platform applications, and we continue to rely on it at every layer of our product.

Uno Platform Team

New Features in 10



XAML

Global Namespaces

Reduce or completely eliminate the xmlns declarations in your project. Declare your own prefixes globally and make your XAML more legible.

Implicit Namespaces

(Preview) Remove the boilerplate xaml at the head of every XAML file for a cleaner, even more legible file. Language Service and editor support will come in the future.

Source Generation

Using Roslyn source generators instead of XAMLC to IL you get faster debug runs, consistent UI on release, and ability to debug generated UI code!

Without Global xmlns

```
<ContentPage xmlns="http://schemas.microsoft.com/dotnet/2021/maui"
             xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
             xmlns:pageModels="clr-namespace:DeveloperBalance.PageModels"
             xmlns:models="clr-namespace:DeveloperBalance.Models"
             xmlns:controls="clr-namespace:DeveloperBalance.Pages.Controls"
             xmlns:pullToRefresh="clr
namespace:Syncfusion.Maui.Toolkit.PullToRefresh;assembly=Syncfusion.Maui.Toolkit"
             xmlns:toolkit="http://schemas.microsoft.com/dotnet/2022/maui/toolkit"
             x:Class="DeveloperBalance.Pages.MainPage"
             x:DataType="pageModels:MainPageModel"
             x:Name="OverviewPage"
             Title="{Binding Today}">
```

Declaring Global xmlns

```
// Global XAML namespace mappings - App namespaces
[assembly: XmlnsDefinition("http://schemas.microsoft.com/dotnet/maui/global", "DeveloperBalance.Pages.Controls")]
[assembly: XmlnsDefinition("http://schemas.microsoft.com/dotnet/maui/global", "DeveloperBalance.PageModels")]
[assembly: XmlnsDefinition("http://schemas.microsoft.com/dotnet/maui/global", "DeveloperBalance.Models")]

// Global XAML namespace mappings - Syncfusion
[assembly: XmlnsDefinition("http://schemas.microsoft.com/dotnet/maui/global",
"Syncfusion.Maui.Toolkit.PullToRefresh", AssemblyName = "Syncfusion.Maui.Toolkit")]

// Global XAML namespace mappings - Community Toolkit
[assembly: XmlnsDefinition("http://schemas.microsoft.com/dotnet/maui/global", "CommunityToolkit.Maui.Behaviors",
AssemblyName = "CommunityToolkit.Maui")]
```

Using Global xmlns

```
<ContentPage xmlns="http://schemas.microsoft.com/dotnet/maui/global"
             xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
             x:Class="DeveloperBalance.Pages.MainPage"
             x:DataType="MainPageModel"
             x:Name="OverviewPage"
             Title="{Binding Today}">
```

Using Global xmlns

```
<Grid>
    <SfPullToRefresh
        IsRefreshing="{Binding IsRefreshing}"
        RefreshCommand="{Binding RefreshCommand}">
        <SfPullToRefresh.PullableContent>
            <ScrollView>
                <VerticalStackLayout Spacing="{StaticResource LayoutSpacing}" Padding="{StaticResource
LayoutPadding}">
                    <Label Text="Task Categories" Style="{StaticResource Title2}"
SemanticProperties.HeadingLevel="Level1"/>
                    <CategoryChart />
                    <Label Text="Projects" Style="{StaticResource Title2}"
SemanticProperties.HeadingLevel="Level1"/>
                    <ScrollView Orientation="Horizontal" Margin="-30,0">
                        <HorizontalStackLayout
                            Spacing="15" Padding="30,0"
                            BindableLayout.ItemsSource="{Binding Projects}">
                            <BindableLayout.ItemTemplate>
                                <DataTemplate x:DataType="Project">
                                    <ProjectCardView WidthRequest="200">
                                        <ProjectCardView.GestureRecognizers>
                                            <TapGestureRecognizer Command="{Binding NavigateToProjectCommand,
Source={RelativeSource AncestorType={x:Type MainPageModel}}, x:DataType=MainPageModel}" CommandParameter="{Binding
.}"/>
                                        </ProjectCardView.GestureRecognizers>
                                    </ProjectCardView>
                                </DataTemplate>
                            </BindableLayout.ItemTemplate>
                        </HorizontalStackLayout>
                    </ScrollView>
```

Implicit Namespaces (Preview)

```
<ContentPage x:Class="DeveloperBalance.Pages.MainPage"
             x:DataType="MainPageModel"
             x:Name="OverviewPage"
             Title="{Binding Today}">

             xmlns:pullToRefresh="clr
namespace:Syncfusion.Maui.Toolkit.PullToRefresh;assembly=Syncfusion.Maui.Toolkit"
             xmlns:toolkit="http://schemas.microsoft.com/dotnet/2022/maui/toolkit"
             x:Class="DeveloperBalance.Pages.MainPage"
             x:DataType="pageModels:MainPageModel"
             x:Name="OverviewPage"
             Title="{Binding Today}">
```

Implicit Namespace (Preview) Opt-in

```
<PropertyGroup>
    <DefineConstants>$(DefineConstants);MauiAllowImplicitXmlnsDeclaration</DefineConstants>
    <EnablePreviewFeatures>true</EnablePreviewFeatures>
</PropertyGroup>
```

XAML Source Generation

Opt-in feature

Faster debug and release performance

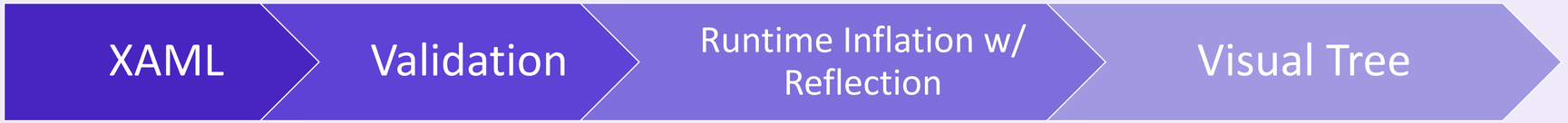
Faster build times (SR1)

Consistent Debug -> Release

Generated code can be debugged

XamlC

Debug



Release



XAML Source Generation

Debug and Release



1000%

Faster Inflation

99%

Less Debug Memory

25%

Faster on Release

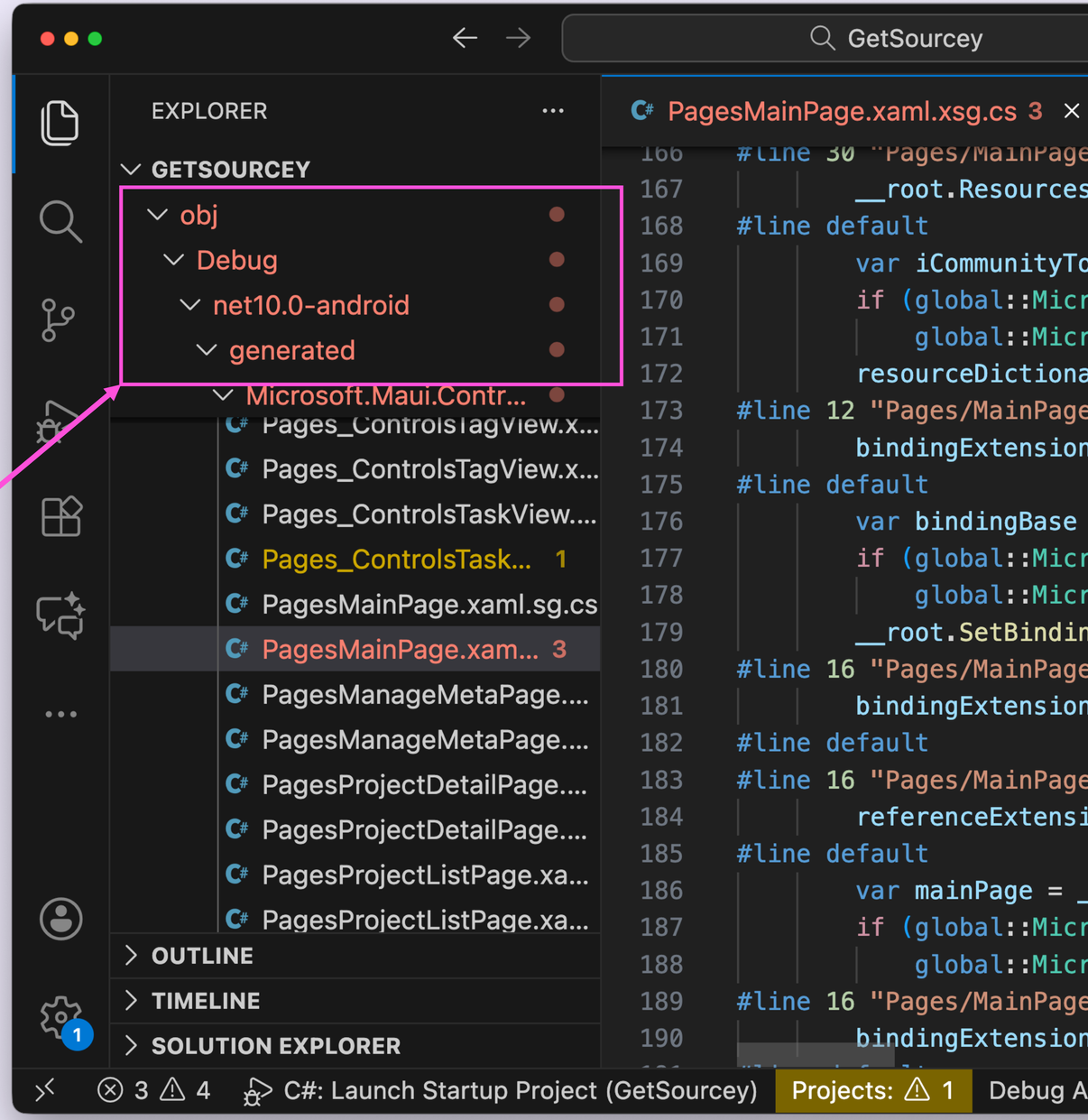
30%

Less App Memory

XAML Source Generation

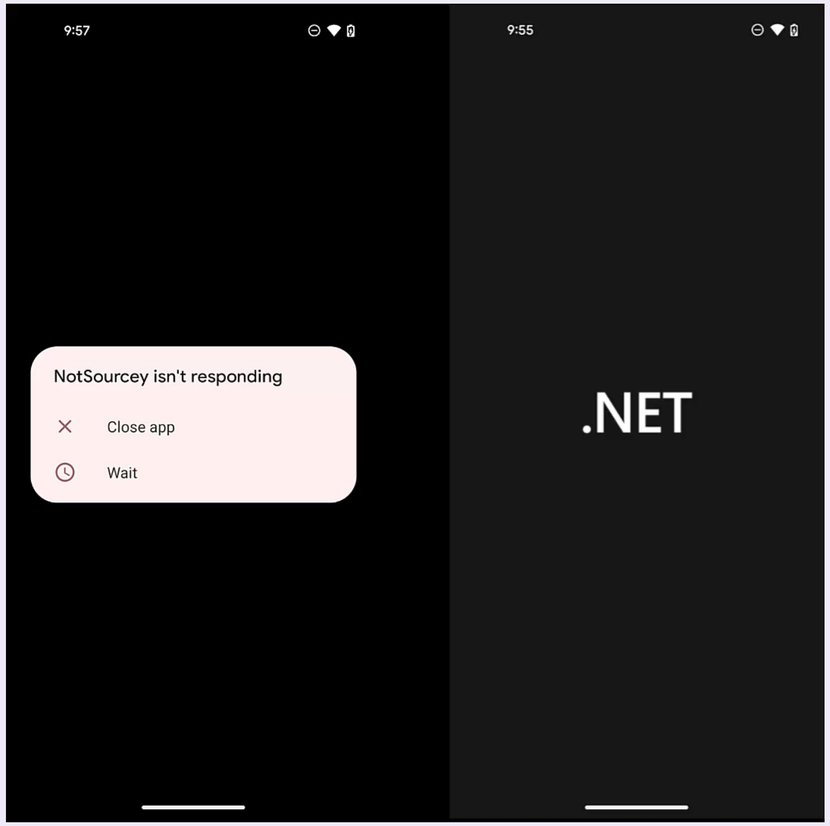
To enable it

```
<PropertyGroup>
  <MauiXamlInflator>SourceGen</MauiXamlInflator>
  <!-- Optional -->
  <EmitCompilerGeneratedFiles>true</EmitCompilerGeneratedFiles>
</PropertyGroup>
```



Performance (Debug) with XAML Source Generation

Splash > Home



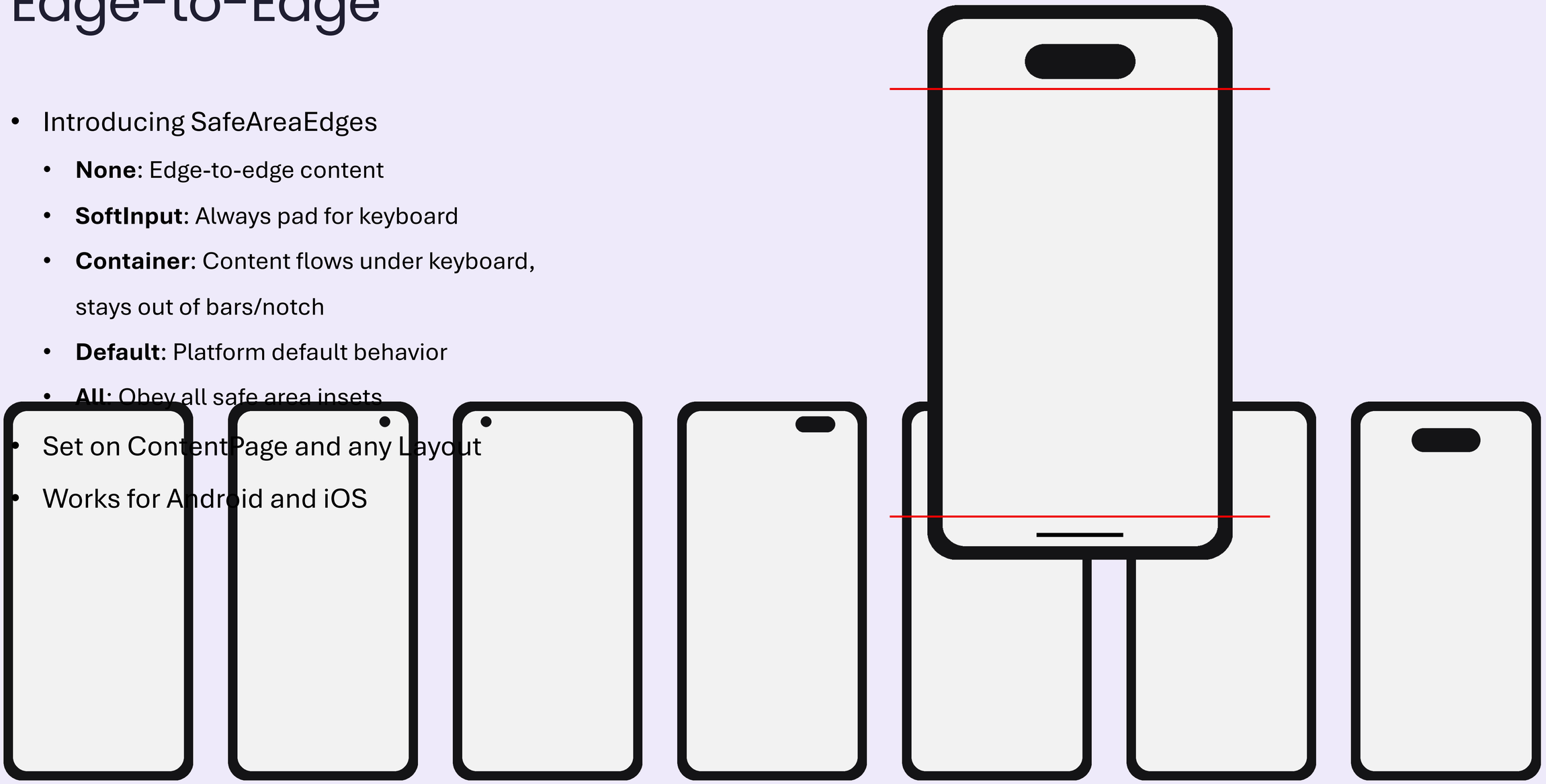
XamlC

Source Gen

Pixel 5, Android 14

Safe Area
Edge-to-Edge

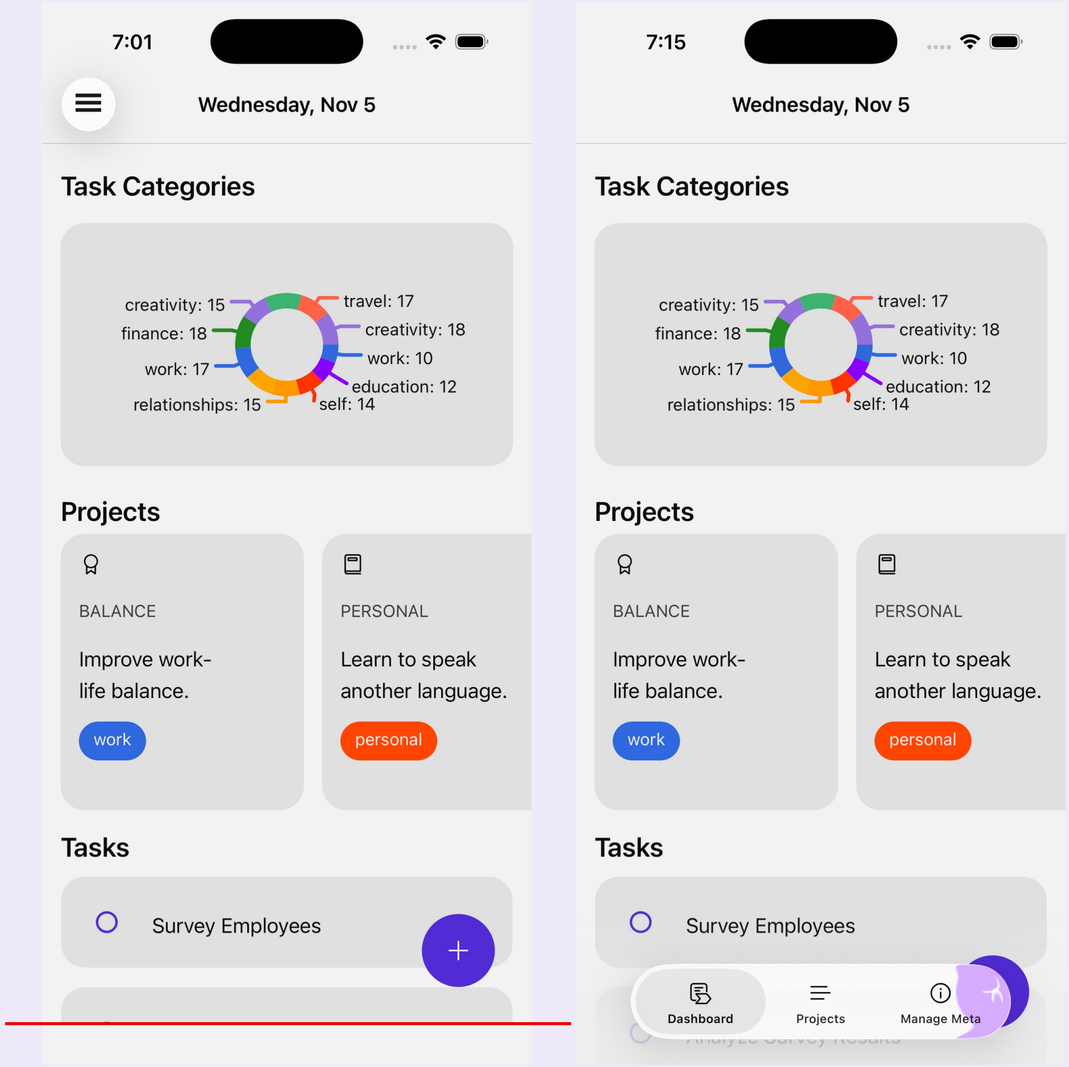
- Introducing SafeAreaEdges
  - **None**: Edge-to-edge content
  - **SoftInput**: Always pad for keyboard
  - **Container**: Content flows under keyboard, stays out of bars/notch
  - **Default**: Platform default behavior
  - **All**: Obey all safe area insets
- Set on ContentPage and any Layout
- Works for Android and iOS



Safe Area
Edge-to-Edge

```
<ContentPage x:Class="GetSourcey.Pages.MainPage"
             x:DataType="pageModels:MainPageModel"
             x:Name="OverviewPage"
             SafeAreaEdges="None"
             Title="{Binding Today}">

    <Grid SafeAreaEdges="None">
        <ScrollView SafeAreaEdges="None">
```



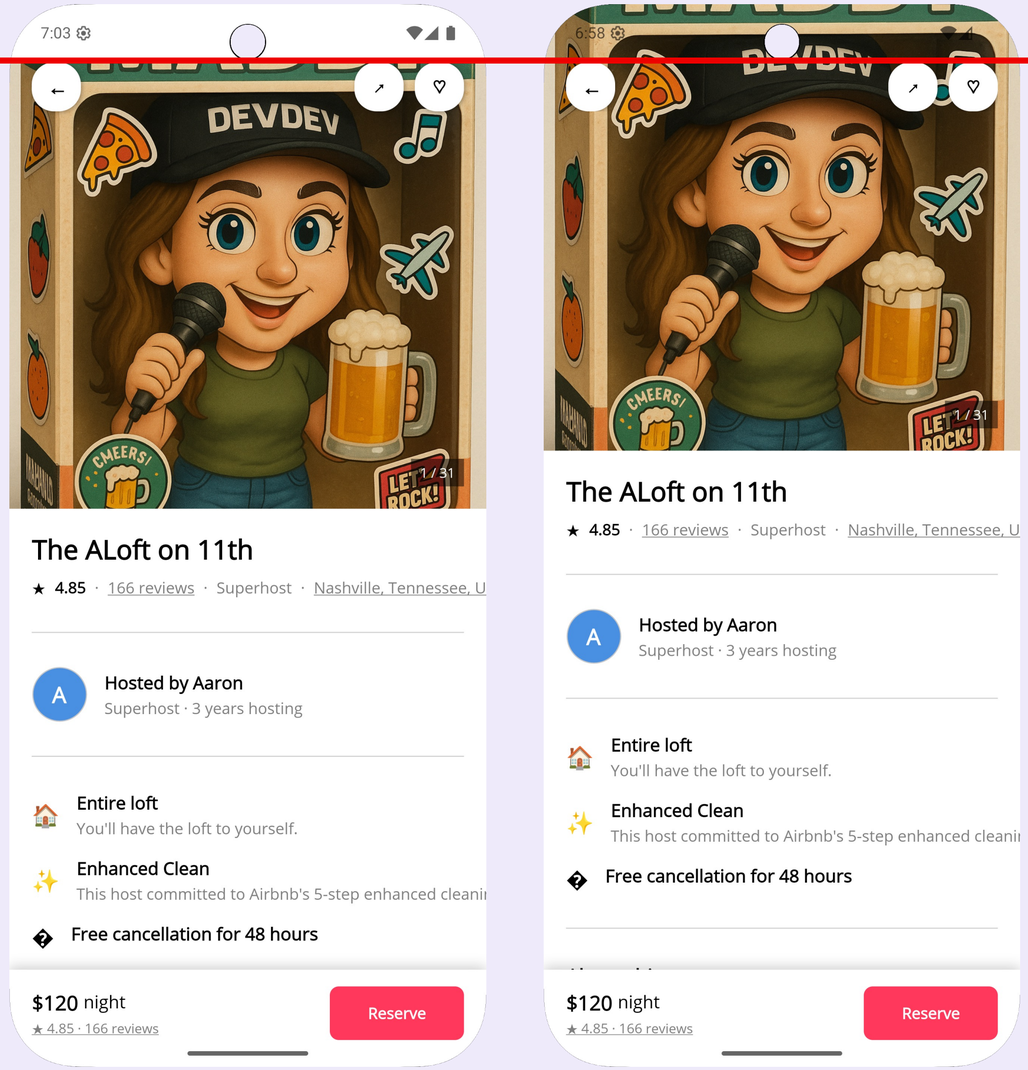
Default

SafeAreaEdges="None"

Safe Area
Edge-to-Edge

```
<ContentPage x:Class="GetSourcey.Pages.MainPage"
             x:DataType="pageModels:MainPageModel"
             x:Name="OverviewPage"
             SafeAreaEdges="None"
             Title="{Binding Today}">

    <Grid SafeAreaEdges="None">
        <ScrollView SafeAreaEdges="None">
```



Default

SafeAreaEdges=“None”

MAUI hybrid apps in .NET 10

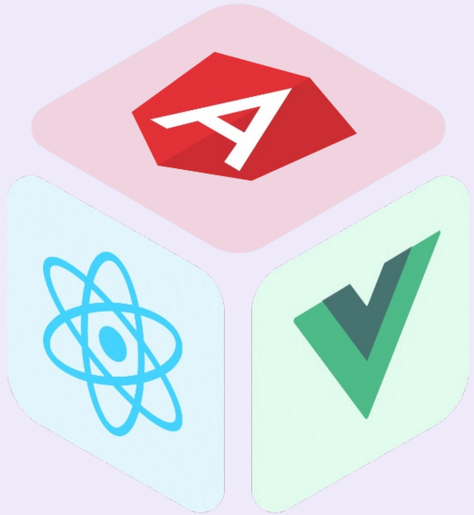
Hybrid apps with Blazor

- Updating solution template for Blazor Hybrid+Web
- Tighter inner loop with Razor and Hot Reload improvements
- More control over the BlazorWebview



Hybrid apps with JavaScript frontends

- Fit-and-finish work, packaging hybridwebview.js
- Better exception handling
- New hooks for intercepting web requests



Hybrid Updates

BlazorWebView

HybridWebView

- InvokeJavaScript
- WebViewInitializing / WebViewInitialized
- Intercept requests with WebResourceRequested

```
<HybridWebView WebResourceRequested="HybridWebView_WebResourceRequested" />

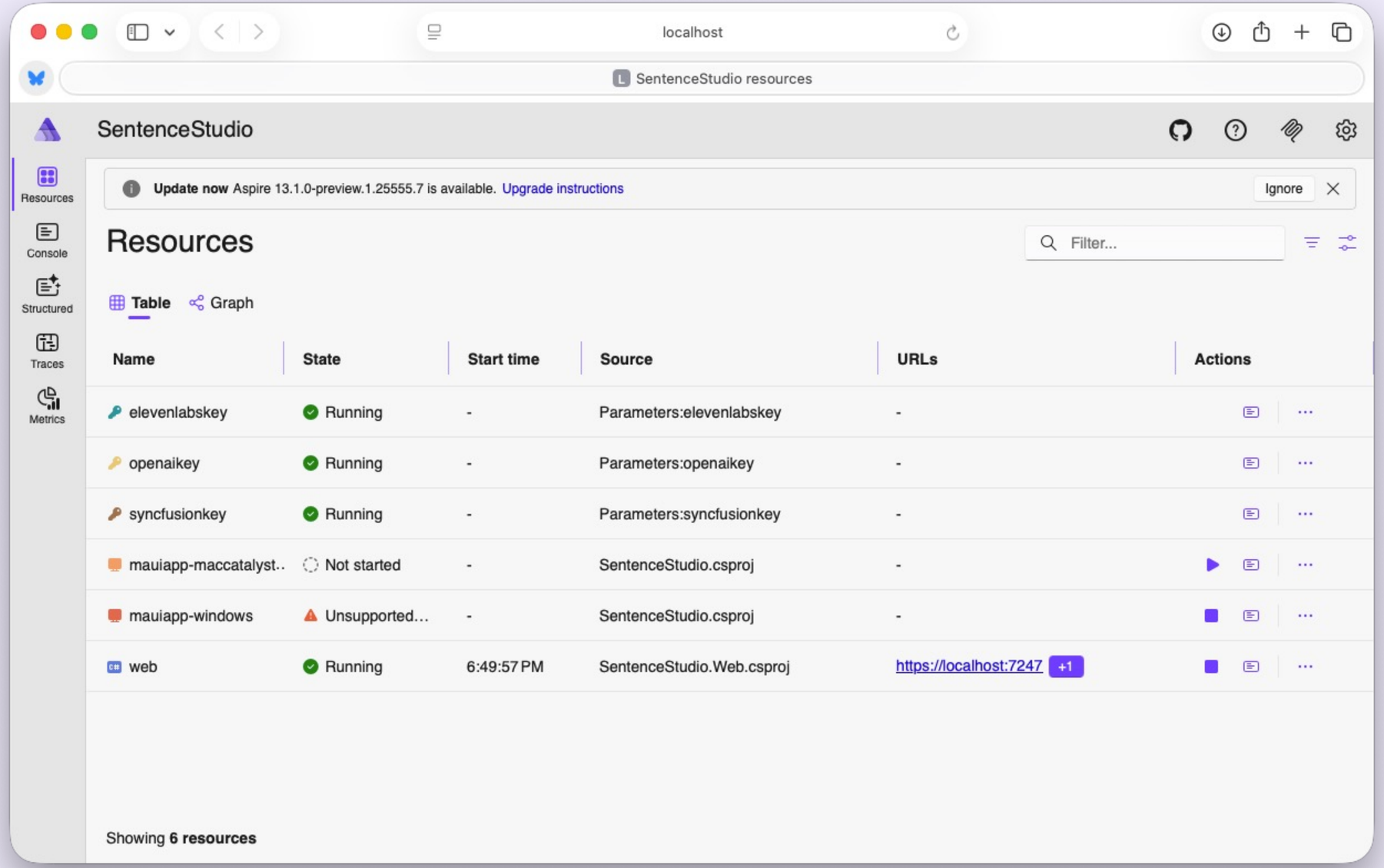
private void HybridWebView_WebResourceRequested(object sender,
 HybridWebViewWebResourceRequestedEventArgs e)
{
  // Only handle requests for the specific image URL
  if (!e.Uri.ToString().Contains("sample-image.png"))
    return;

  // Prevent the default behavior of the web view
  e.Handled = true;

  // Return the stream or task of stream that contains the content
  // NOTE: the method is NOT awaited, the WebView will continue to load the content
  e.SetResponse(200, "OK", "image/png", GetStreamAsync());
}
```

Aspire
Orchestration

- Define your stack
  - Front-end
  - Database
  - APIs
  - Containers
- Configuration
- Observability
- Deployment



aspire.dev

Aspire
Orchestration

- Define your stack
  - Front-end
  - Database
  - APIs
  - Containers
- Configuration
- Observability
- Deployment

```
using Projects;

var builder = DistributedApplication.CreateBuilder(args);

var openaikey = builder.AddParameter("openaikey", secret: true);
var syncfusionkey = builder.AddParameter("syncfusionkey", secret: true);
var elevenlabskey = builder.AddParameter("elevenlabskey", secret: true);

var webapi = builder.AddProject<SentenceStudio_Web>("web");

var mauiapp = builder.AddMauiProject("mauiapp", "../SentenceStudio/SentenceStudio.csproj");

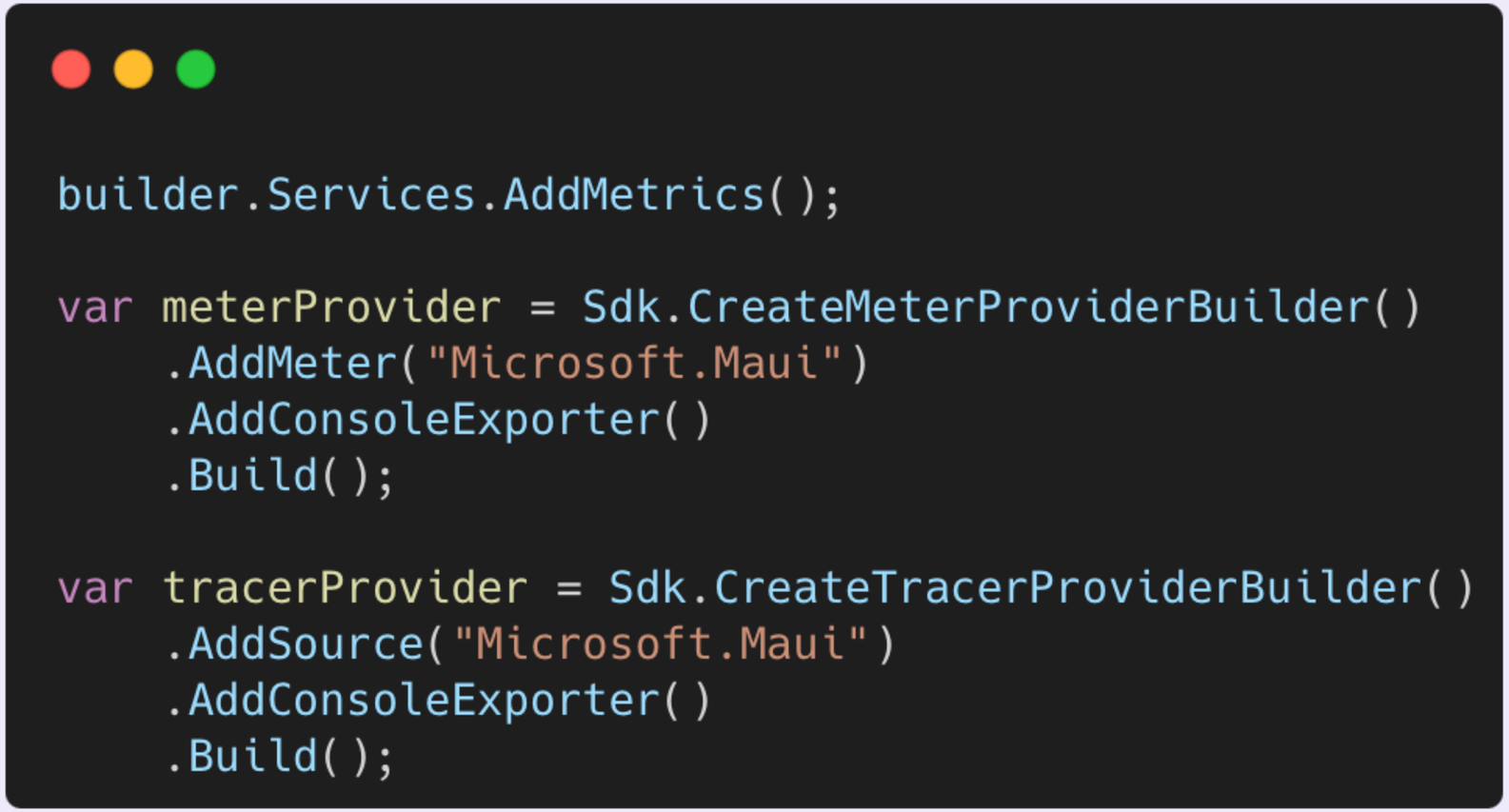
mauiapp.AddMacCatalystDevice()
    .WithEnvironment("SyncfusionKey", syncfusionkey)
    .WithEnvironment("AI__OpenAI__ApiKey", openaikey)
    .WithEnvironment("ElevenLabsKey", elevenlabskey)
    .WithReference(webapi);

mauiapp.AddWindowsDevice()
    .WithEnvironment("SyncfusionKey", syncfusionkey)
    .WithEnvironment("AI__OpenAI__ApiKey", openaikey)
    .WithEnvironment("ElevenLabsKey", elevenlabskey)
    .WithReference(webapi);

builder.Build().Run();
```

aspire.dev

Diagnostics
Monitor your perf



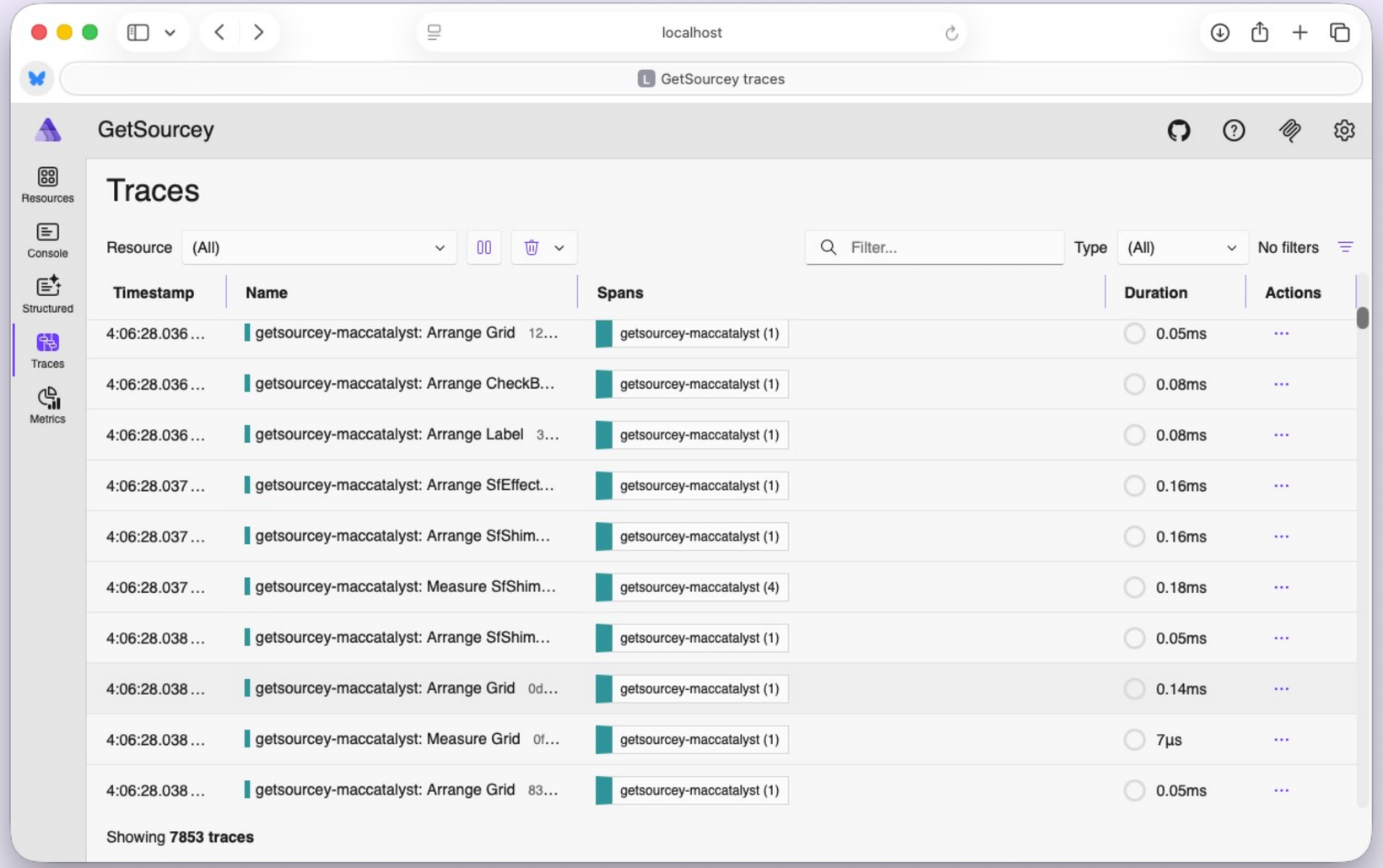
```
builder.Services.AddMetrics();

var meterProvider = Sdk.CreateMeterProviderBuilder()
    .AddMeter("Microsoft.Maui")
    .AddConsoleExporter()
    .Build();

var tracerProvider = Sdk.CreateTracerProviderBuilder()
    .AddSource("Microsoft.Maui")
    .AddConsoleExporter()
    .Build();
```

Diagnostics

Monitor your perf



Diagnostics
Monitor your perf

```
builder.Services.AddMetrics();

var meterProvider = Sdk.CreateMeterProviderBuilder()
    .AddMeter("Microsoft.Maui")
    .AddConsoleExporter()
    .Build();

var tracerProvider = Sdk.CreateTracerProviderBuilder()
    .AddSource("Microsoft.Maui")
    .AddProcessor(new SimpleConsoleProcessor())
    .Build();


public class SimpleConsoleProcessor : BaseProcessor<Activity>
{
    public override void OnEnd(Activity activity)
    {
        Console.WriteLine($"{activity.DisplayName}: {activity.Duration.TotalMilliseconds}ms");
    }
}
```

Diagnostics
Monitor your perf

```
builder.Services.AddMetrics();

var meterProvider = Sdk.CreateMeterProviderBuilder()
    .AddMeter("Microsoft.Maui")
    .AddConsoleExporter()
    .Build();

var tracerProvider = Sdk.CreateTracerProviderBuilder()
    .AddSource("Microsoft.Maui")
    .AddProcessor(new SimpleConsoleProcessor())
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public class SimpleConsoleProcessor : BaseProcessor<Activity>
{
    public override void OnEnd(Activity activity)
    {
        Console.WriteLine($"{activity.DisplayName}: {activity.Duration.TotalMilliseconds}ms");
    }
}
```

```
[DOTNET] Measure Image: 0.0883ms
[DOTNET] Measure Label: 0.0762ms
[DOTNET] Measure Grid: 1.0008ms
[DOTNET] Arrange Image: 0.2328ms
[DOTNET] Arrange Label: 0.3507ms
[DOTNET] Arrange Grid: 1.3091ms
[DOTNET] Measure Image: 0.1181ms
[DOTNET] Measure Label: 0.0671ms
[DOTNET] Measure Grid: 0.8641ms
[DOTNET] Arrange Image: 0.2263ms
[DOTNET] Arrange Label: 0.352ms
[DOTNET] Arrange Grid: 1.2846ms
[DOTNET] Measure Image: 0.0692ms
[DOTNET] Measure Label: 0.0873ms
[DOTNET] Measure Grid: 0.8073ms
[DOTNET] Arrange Image: 0.2236ms
[DOTNET] Arrange Label: 0.3403ms
```

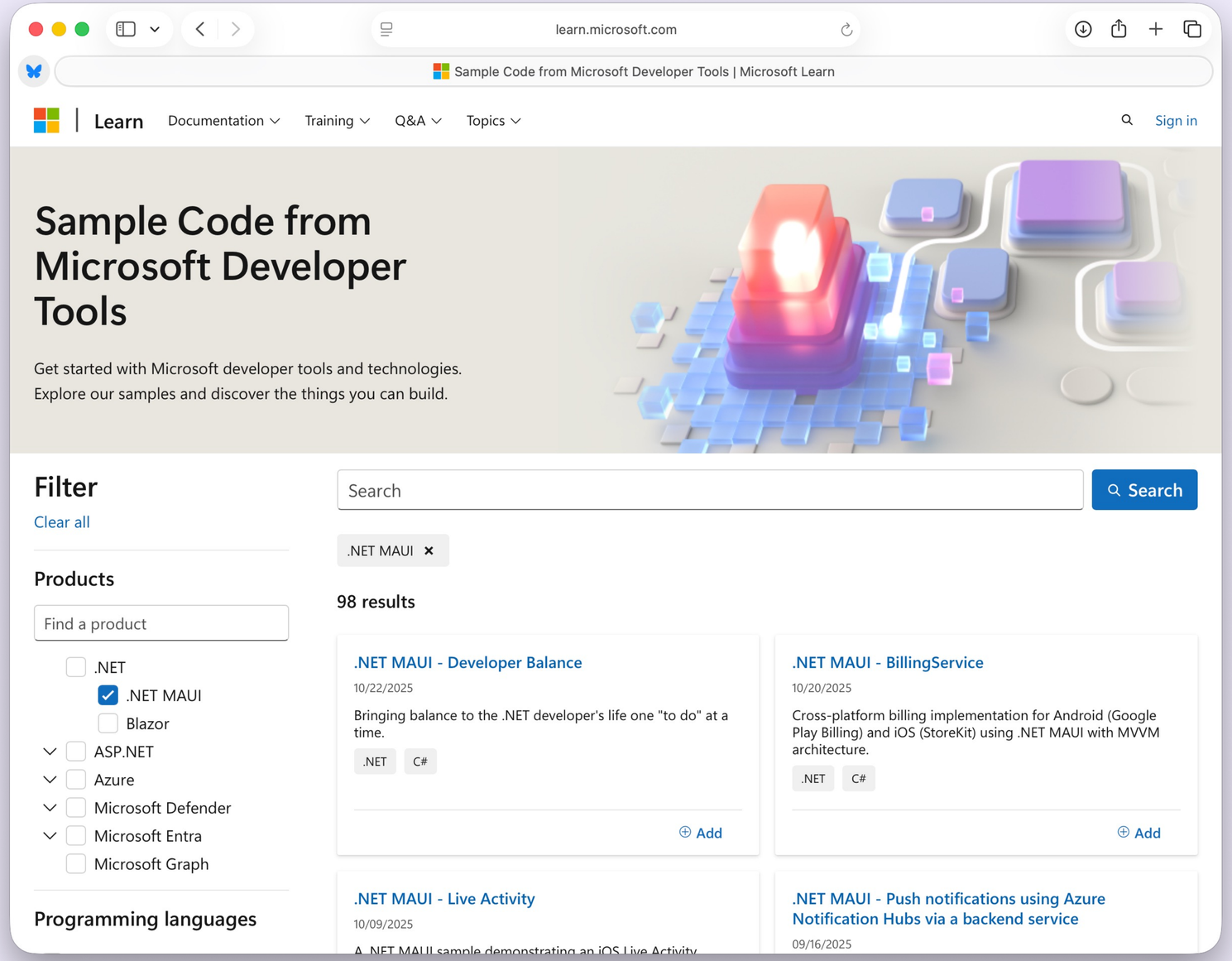
And more

aka.ms/whats-new-10

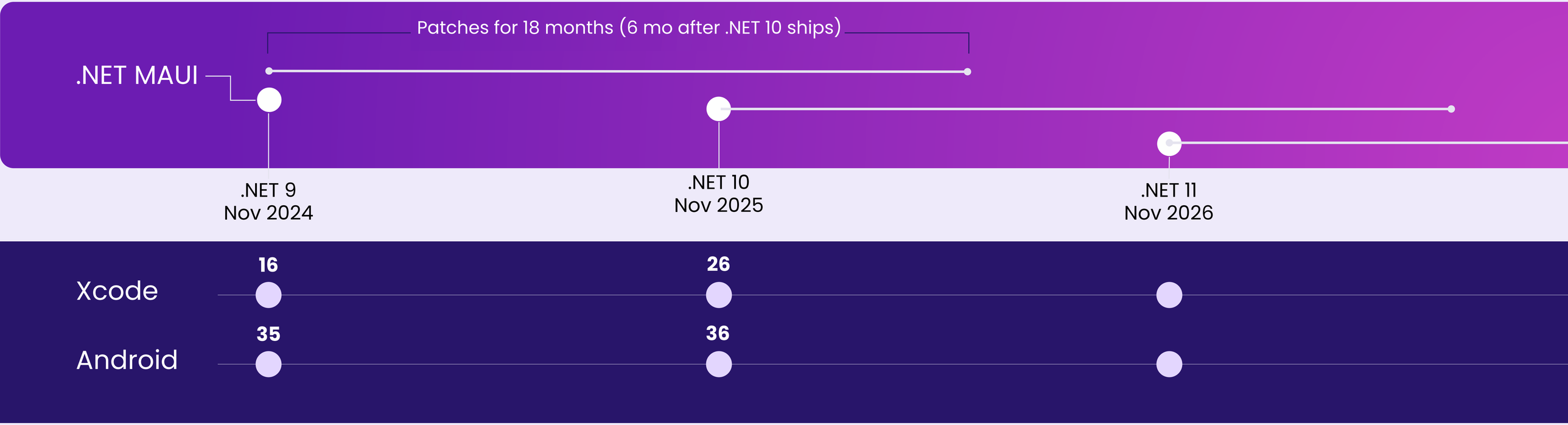
- Accessibility
- Animation
- CollectionView
- CarouselView
- Editor
- Entry
- Picker
- RefreshView
- SearchBar
- Switch
- Vibration & HapticFeedback
- Window
- Switch
- Modals
- GeoLocation
- MediaPicker
- TextToSpeech
- WebAuthenticator
- ToolBarItems
- Messaging
- Pop-ups
- Markup Extensions
- Font Images
- Gesture recognizers
- Page
- Shadows
- CoreCLR
- Native AOT
- Xcode 26
- Android API 36

Resources

- aka.ms/whats-new-in-10
- aka.ms/maui-docs
- aka.ms/maui-samples
- aka.ms/maui-10-upgrade



Release schedule, support & patching



aka.ms/maui-support-policy

.NET MAUI at .NET Conf 2025

Tuesday @ 4:30pm PST

.NET Diagnostic Tooling with AI
Jonathan Peppers

Wednesday @ 5:30pm PST

Migrating from Xamarin.Forms to NET. MAUI: The Hard Parts
Iris Classon

Thursday @ 12pm PST

Uno Platform. Now with AI
Jérôme Laban

Thursday @ 3:30pm PST

Ship Faster with .NET MAUI
Paul Usher

Thursday @ 4:30pm PST

Community Toolkit Roundup
Michael Hawker, Gerald Versluis,
& Sergio Pedri

Thursday @ 11pm PST

Architecting an AI-Powered Sales Dashboard with .NET MAUI & Azure OpenAI
Shriram Sankaran

Thank you

